

Automated School IT Helpdesk Ticket Classification in ServiceNow

PHASE 3: PROJECT DESIGN PHASE

The architectural design defines the custom database table schema, role-based security model, form field structures, and choice list dependency mappings.

- **Database Architecture:** A custom table in the Global scope named Incident WorkFlow (backend name: u_incident_workflow) stores all incoming ticket records. The table is configured with an INC auto-number prefix (configured at start 1000 with 7 digits in the live instance, e.g., INC0001001, and 500 with 5 digits in baseline specification, e.g., INC00500) and is secured via role-based access control under the u_incident_workflow_user role.

Form Fields & Choice Dependencies: The table layout incorporates standard incident management fields and enforces strict parent-child choice dependencies:

- **Number:** Auto Number field (Prefix: INC, Starting number: 1000 / 500, Number of digits: 7 / 5).
- **Caller:** Reference field (u_caller) pointing to the core User table (sys_user).
- **Short Description & Description:** String fields (u_string_4, u_string_5) capturing summary and detailed issue text.
- **State:** Choice list (u_choice_6) containing New, In progress, On hold, Resolved, and Closed.
- **Assigned Group & Assigned to:** Reference fields (u_reference_7, u_reference_8) pointing to sys_user_group and sys_user.
- **Category:** Choice list (u_choice_2) with Network (network), Hardware (hardware), Access (access), and Performance (performance).
- **Subcategory:** Dependent choice list (u_choice_3) with Wi-Fi (wi-fi), Projector (projector), Forgot Password (forgot password), and Slow Computer (slow compmer / slow_computer).

Table 1: Incident WorkFlow (u_incident_workflow) Schema & Choice Architecture

Field Label	Backend Column	Data Type / Target	Configuration & Choice Values
Number	number	Auto Number	Prefix: INC Start: 1000 (7 digits, e.g., INC0001004) / Spec: 500 (5 digits)
Caller	u_caller	Reference (sys_user)	Identifies the teacher or student submitting the ticket
Category	u_choice_2	Choice	Network, Hardware, Access, Performance
Subcategory	u_choice_3	Dependent Choice	Wi-Fi, Projector, Forgot Password, Slow Compter (Slow Computer)

Short Description	u_string_4	String (Max 40)	Evaluated by Flow Designer for classification keywords
Description	u_string_5	String (Max 40)	Detailed explanation of the reported IT issue
State	u_choice_6	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	u_reference_7	Reference (sys_user_group)	Links to support group for ticket routing
Assigned to	u_reference_8	Reference (sys_user)	Links to individual IT support agent

Table 2: Parent-Child Dictionary Dependency & Keyword Classification Rules

Short Description Keyword(s)	Parent Category Assigned	Dependent Subcategory Assigned	Flow Designer Branch
Contains 'Wi-Fi' OR 'Network'	Network	Wi-Fi	Branch 1: If (SD is Wifi or Network)
Contains 'Projector'	Hardware	Projector	Branch 2: Else If (SD is Projector or Hardware)
Contains 'Forgot password'	Access	Forgot Password	Branch 3: Else If (SD is Forgot password)
Contains 'Slow Computer'	Performance	Slow Compter (Slow Computer)	Branch 4: Else If (SD is Performance)

Complete Incident Workflow (u_incident_workflow) Form Design Layout with All 9 Configured Fields